Impact of short-rotation coppice with poplar and willow on soil physical properties

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Abstract

As the land area of short-rotation coppice (SRC) increases, their soil physical impacts have to be evaluated. The objective of this study was to detect the effects of long-term SRC with poplar and willow on the vertical distribution of soil physical properties (bulk density, water retention, penetration resistance) and on solute transport patterns. An 18-year-old SRC located in northeastern Germany was compared to an adjacent continuous arable cropping system by means of soil sampling, penetrologger measurements and dye tracer experiments. The topsoil's bulk density was significantly lower under SRC than under cropland. This effect was especially pronounced in the uppermost 10 cm, where also the air capacity and the plant-available water content were higher under SRC. The penetration resistance in 25–50 cm depth was reduced under SRC compared to the cropland, indicating a loosening of the plough pan. Dye tracer experiments showed that the importance of preferential flow was higher under SRC due to tree root channels and an increasing colonisation with invertebrates. SRC has ecologically advantageous effects on soil physical properties of the topsoil, however, combined with an enhanced risk of preferential solute transport upon application of agrochemicals.

Key words: bulk density / dye tracer / fast-growing trees / penetration resistance / Salicaceae / water retention

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1 Introduction

Short-rotation coppice (SRC) with poplar (*Populus* ssp.) and willow (*Salix* ssp.) is widely used to produce biomass for energy generation (*Berhongaray* et al., 2017). Short rotation coppice is typically established on land that was formerly under conventional arable use, and is characterized by a no-till perennial use (10–25 years) and harvested every 2–6 years (*Baum* et al., 2009; *Dimitriou* et al., 2009). The advantages of poplar and willow for SRC are their high energy efficiency and large biomass production potentials compared to conventional arable crops, while requirements for fertilisation, pest management and tillage are low (*Aylott* et al., 2008; *Dimitriou* et al., 2009). Possible drawbacks of SRC are large water requirements and their possible impact on local water resources (*Petzold* et al., 2011; *Hartwich* et al., 2016) as well as the potential to threaten the water quality, depending on the management practices especially in the phases of establishment and final removal of the trees (*Goodlass* et al., 2007; *Nisbet* et al., 2011).

The area covered with SRC has increased in many European countries as a result of a growing demand for energy wood and favourable conditions for subsidies in the European Union, *e.g*., in Sweden (*Hoffmann* and *Weih*, 2005), Finland (*Wall* and *Heiskanen*, 2003), Great Britain (*Rowe* et al., 2009), Ireland (*Murphy* et al., 2014), and Germany (*Murach* et al., 2008). As the land area under SRC increases, more detailed information on the potential impacts of SRC on the soil is needed. So far, investigations of the impact of poplar or willow SRC on soil focus mainly on chemical or biological characteristics, e.g. carbon sequestration potential (*Berhongaray* et al., 2017), microbial biomass, colonization and activities in the soil (*Mao* and *Zeng*, 2010), and nutrient dynamics (*Ens* et al., 2013).

Little is known, however, about the impact of SRC on the physical properties of soils. The soil physical properties are crucial for plant growth, since they govern the amounts of water and air available for the plants and thus influence root and biomass growth. Available studies are restricted to SRC with young stand ages and concentrate on the topsoil. Low bulk densities in the topsoil have been reported as a consequence of SRC site preparation, which subsequently increased in the course of nine years due to natural compaction (*Makeschin*, 1994). Analogously, *Schmitt* et al. (2010) found increased bulk densities and decreased saturated hydraulic conductivities in the uppermost layer of the topsoil after four years of poplar and willow cultivation, while no changes were observed in the deeper soil layers. *Rytter* (2016), in contrast, did not observe any significant differences in the bulk density down to 30 cm soil depth between the planting year and five

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years later. Investigations on the long-term effects of SRC on soil physical properties are missing so far.

The soil physical properties also govern the flow of water and associated solute transport through the soil profile. Preferential flow, defined as water and solutes moving along certain pathways while bypassing the biologically active soil matrix (*Hendrickx* and *Flury*, 2001), might promote a fast transport of solutes through the vadose zone to the groundwater and reduce the possibility for degradation processes. Dye tracing allows visualizing the water flow pathways in unsaturated soil profiles. It is thus a suitable method to assess whether and how long-term SRC affects water flow and solute transport.

Increased nitrate-N concentrations in drainage waters have been observed upon initial establishment of a SRC and upon final removal of the trees (*Goodlass* et al., 2007). After the establishment peak, N concentrations gradually decreased (*Schmidt-Walter* and *Lamersdorf*, 2012; *Díaz-Pinés* et al., 2017) and were lower than under conventional arable cropping during the productive harvest phase, so that N losses are expected to be smaller under SRC than conventional cropping in the lifespan of a typical plantation (*Goodlass* et al., 2007). A good management of the SRC can thus prevent nitrate leaching (*Aronsson* et al., 2000; *Schmidt-Walter* and *Lamersdorf*, 2012). Nutrient supply in SRC established on former arable land is usually sufficient, at least during the initial rotations (*Petzold* et al., 2011). On marginal-yield sites, such as recultivated mining areas, however, fertilisation might be necessary (*Röhle* et al., 2008).

The objectives of this study were (1) to detect the impact of long-term SRC with poplar and willow on the vertical distribution of soil physical properties (bulk density, water retention, penetration resistance, water repellency) in comparison to continuous conventional cropping and (2) to evaluate their impact on solute transport patterns.

2 Material and methods

2.1 Study site

The study plots are located in the federal state of Mecklenburg-Western Pomerania, northeastern Germany, near the village of Gülzow (53°49′20″ N, 12°04′05″ E). The site lies within Pleistocene lowlands, the parent material is glacial till. The mean annual temperature is 9.1°C, the mean annual precipitation is 577 mm (1988–2017; data provided by Landesforschungsanstalt für Landwirtschaft und Fischerei Mecklenburg-Vorpommern, Institut für Pflanzenproduktion und Betriebswirtschaft, Gülzow).

The SRC was established in 1993 on a former arable field sized 1.4 ha. Twenty-eight clones of poplar and willow were planted in total, two of which are included in the present investigation. The plot size for each clone is 4.5 m × 30 m and the plant spacing 1.5 m × 0.5 m, resulting in three tree rows per plot and 13,300 trees ha^{-1}. The above-ground biomass is harvested manually in rotation periods of 3-years in the months of January/February, most recently in 2014 and 2011. The stools are left in the soil for regrowth. Neither fertilizers, pesticides nor irrigation water have been applied to the SRC plantation since its establishment.

The adjacent cropland reference site is regularly ploughed (to depths of 20–22 cm upon seeding and 28–30 cm in autumn) and has been cropped with maize (2015), winter barley followed by catch crops like buck wheat, phacelia, persian clover, spring oat (2014), lupins (2013), winter wheat (2012), Italian Ryegrass (2011), summer barley (2010), and winter grain, grass, maize and legume in the years before. For details on management and fertilization, see *Kahle* et al. (2010). Previous investigations at the site Gülzow showed that the organic carbon content and total mycorrhizal fungi colonization in the topsoil were already increased after only six years of SRC (*Kahle* et al., 2005). Short rotation coppice significantly influenced the vertical distribution of carbon and nitrogen in the soil profile (*Kahle* et al., 2010), and the removal of the trees caused a rapid redistribution of the organic matter and a loss of aggregate stability in the topsoil (*Kahle* et al., 2013).

2.2 Soil profiles

In June 2011, 18 years after establishment of the SRC, four soil profiles with a depth of 1 m were prepared, two in the SRC and two reference plots in the adjacent cropland with continuous conventional cropping. The two SRC plots were planted with poplar (*P. trichocarpa* × *P. deltoides* cv. Beaupré, SRC-P) and willow (*S. caprea* × *S. viminalis*, clone 6 of the Waldsieversdorf collection, SRC-W), and were 130 m apart from each other. The soil profiles were located in the middle of the respective treatment plot. The two profiles located in the cropland (Cropland-P, Cropland-W) had a distance of about 15 m from plots SRC-P and SRC-W, respectively.

Soil texture in all profiles was mainly sandy loam (Tab. 1). Soil types were Haplic Cambisol (Calcaric) in the profiles SRC-P and Cropland-P, and Stagnic Cambisol in the profiles SRC-W and Cropland-W. The different soil types developed according to their position within the slightly sloped (< 3%) SRC plantation: Profiles SRC-P and Cropland-P were located in a higher, profiles SRC-W and Cropland-W in a lower topographical position. As a consequence, the marly till with carbonate contents of 1.2–12.5% was found starting at about 70 cm depth in profiles SRC-P and Cropland-P, while carbonate contents were below 0.1% in the entire profiles SRC-W and Cropland-W (Tab. 1). The paired profiles (SRC-P and Cropland-P; SRC-W and Cropland-W) were largely similar with respect to soil horizons, horizon thickness, texture and carbonate content, rendering a comparison of the two land uses possible. Only in the subsoil (below 35 cm depth at SRC-P, 70–80 cm depth at SRC-W), clay contents differed.

2.3 Laboratory measurements

Disturbed and undisturbed soil samples were taken in five depths (*cf*. Tab. 1). The disturbed samples, which were composites of five individual samples distributed over the profile wall, were air-dried and sieved to < 2 mm. Particle size distribution was determined using a combined sieve and pipette

Table 1: Particle size distribution (n = 1), content of $CaCO_3$ (mean, n = 2) and soil organic carbon (SOC) (mean, n = 2) in the soil under short rotation coppice with poplar (SRC-P) and willow (SRC-W), and at paired reference sites under cropland (Cropland-P and Cropland-W).

Plot	Depth (cm)	Clay (%)	Silt (%)	Sand (%)	$CaCO_3$ (%)	SOC (%)
SRC-P	0–10	9.0	29.4	61.6	0.09	1.66
	10–20	10.3	28.1	61.6	0.09	0.76
	20–30	10.9	26.4	62.7	0.07	0.64
	35–45	12.2	30.1	57.7	0.14	0.50
	70–76	9.7	34.3	56.0	12.5	n.d.
Cropland-P	0–10	8.6	25.3	66.1	0.05	0.94
	10–20	8.1	26.5	65.4	0.05	0.95
	20–30	8.7	24.1	67.2	0.05	0.83
	35–45	24.2	31.4	44.4	0.08	0.28
	70–80	22.6	28.7	48.8	1.20	0.22
SRC-W	0–10	7.4	25.5	67.1	0.09	1.19
	10–20	8.6	26.2	65.2	0.09	0.54
	20–30	7.0	28.2	64.8	0.09	0.55
	35–45	7.7	25.9	66.4	0.05	0.55
	70–80	14.6	29.3	56.1	0.05	0.23
Cropland-W	0–10	7.7	24.7	67.6	0.09	0.75
	10–20	10.5	21.9	67.6	0.05	0.76
	20–30	7.1	25.4	67.5	0.05	0.58
	35–45	7.1	27.8	65.1	0.05	0.30
	70–80	7.9	26.7	65.4	0.09	0.17

method (*Hartge* and *Horn*, 1999). The total carbon content (C_t) was measured in duplicates by dry combustion using a Vario EL analyzer (Elementar Analysesysteme GmbH, Hanau, Germany). The inorganic soil carbon (C_{inorg}) was measured by dissolution with HCl and volumetric CO_2 determination. The soil organic carbon content (SOC) was determined by substracting C_{inorg} from C_t.

Undisturbed soil samples (250 cm^3) were taken in the centre of each soil horizon with eight replicates. The water retention parameters were determined using ceramic suction plates at pF 1.8 and 2.48 (corresponding to 60 and 300 hPa; 8 replications) and a pressure membrane apparatus at pF 4.2 (corresponding to 15,000 hPa, 3 replications) (*Hartge* and *Horn*, 1999). Subsequently, bulk density was determined using the core method (*Grossman* and *Reinsch*, 2002). The particle density was assumed as 2.65 g cm^{-3} for calculating the total porosity TP. The air capacity was calculated as $TP - \Theta_{pF1.8}$; the plant-available water as $\Theta_{pF1.8} - \Theta_{pF4.2}$, where Θ is the volumetric water content at the respective pressure head (%). The determined soil properties were evaluated using the German soil classification manual (*AG Boden*, 2005).

Additional undisturbed soil samples (250 cm^3) were taken for analysis of water repellency in 0–5, 5–10, and 10–15 cm depth with two replicates for each of the four treatments. Soil samples were saturated from below for 7 d and then left to dry under laboratory conditions, similar to the procedure described in *Hewelke* et al. (2014). A Water Drop Penetration Time (WDPT) test was performed every 2–3 d, and finally after drying for 24 h at 40°C and 105°C. Three droplets (50 μL) of deionized water were positioned on the surface of each sample, and the time for disappearance of the droplets was measured and classified according to *Dekker* and *Ritsema* (2000).

2.4 Dye tracer experiments

Dye tracers are commonly used to visualise and identify water flow pathways, and to assess the soil's risk exposure to preferential solute transport (*Flury* et al., 1994). In each of the four plots, a dye tracer experiment was conducted in June 2011. Brilliant Blue was used as dye, since it is a nontoxic dye with high visibility. A metal frame sized 1 m × 1 m was inserted into the soil. Fifty litre of water (corresponding to a depth equivalent of 5 cm) containing 4 g L^{-1} Brilliant Blue were poured onto a plastic sheet within the frame, which was then carefully removed to ensure uniform flooding conditions. The following day, 7 vertical profile sections sized 1m × 1 m at intervals of 10 cm were prepared at each experimental plot and photographed.

A geometric correction of each image was performed by help of a metric frame. The pixel size was set to 0.05 cm. Subsequently, a band ratio of the blue and red band was calculated, based on which a threshold was determined visually to classify the image into stained and unstained pixels [for a detailed description of the procedure, see *Janssen* and *Lennartz* (2008)]. The proportion of stained pixels in each row of pixels was then calculated to derive the dye coverage (DC) with depth and the average dye coverage (ADC) over the entire image. Subsequently, several indices for preferential flow were calculated: the uniform infiltration depth (UID) is defined as the depth at which the dye coverage decreases below 80% (*van Schaik*, 2009). The preferential flow fraction is calculated as (*van Schaik*, 2009):

$$PFF = 100 \times (1 - UID \times Profile\ Width / Total\ Stained\ Area). \tag{1}$$

The length index (LI) is the sum of the absolute differences between DC in consequent rows over the profile and reflects the degree of heterogeneity of the dye pattern (*Bargués Tobella* et al., 2014). For images of average dye distribution, all seven binary images of one plot were superimposed.

The results showed that one part of the dye tracer plots in the cropland (the part closest to the SRC) was located on a hidden machinery track. The dye tracer tests were therefore repeated in those plots in April 2015. The pits for soil sampling in 2011 were located on the other side of the dye tracer plots and were not affected by the vehicle track. Soil moisture conditions were similar at both dates in the topsoil (0–30 cm). Average values amounted to 23 and 22% in 2015 (Cropland-P and Cropland-W) compared to 21 and 21% in 2011. In the deeper soil layers, volumetric water contents at Cropland-P were lower in 2015 (average 11%) than in 2011 (22%), while it was *vice versa* at Cropland-W (25% in 2015 and 15% in 2011).

2.5 Penetration resistance

The penetration resistance was measured with a penetrologger (Eijkelkamp) down to 80 cm depth in April 2015 (cone base area 1 cm^2, cone angle 60°, rate of penetration 2 cm s^{-1}, depth resolution 1 cm). Measurements were taken at 15 locations in each plot, along three parallel transects (distance to the trees: 0.75 m). Five locations were chosen randomly within each transect. If roots or stones impeded a further penetration of the cone, the measurement was discarded and repeated at another location.

Based on the penetration resistance, mechanical stresses of the soils were assessed following *Hartge* and *Bachmann* (2004) and *Horn* et al. (2007). It was assumed that the soil was in a normal-compacted state at 80 cm depth and that penetration resistance increases linearly with depth in the normal-compacted state. The hydrostatic stress distribution in the normal-compacted state was then constructed by linearly connecting the average measured penetration resistance in 80 cm depth with the origin, and the stresses at rest coefficient K_0 was calculated for each depth by dividing the average measured penetration resistance by the assumed hydrostatic stress.

2.6 Statistical analyses

The soil properties were analysed using mean comparisons of data with homogenous variance for the two different management treatments (SRC and cropland) for paired plots within the same depth using Student's t-test. ($p < 0.05$ and $p < 0.01$).

3 Results

3.1 Soil physical properties

The soil bulk density differed significantly between the 18-year-old SRC and the cropland in the upper 20 cm of the soil profile. The difference was most pronounced in the uppermost soil layer (0–10 cm), where the average bulk density amounted to 1.38 and 1.35 g cm^{-3} in the SRC [classified as 'low' according to *AG Boden* (2005)] and to 1.58 and 1.65 g cm^{-3} in the cropland (classified as 'medium' to 'high') (Fig. 1). Below 20 cm depth, there were no significant differences between SRC-P and Cropland-P.

Complementary to the bulk density, the total porosity was significantly higher under SRC than under cropland down to 20 cm depth (Tab. 2). The water retention capacity at pF 1.8 and pF 2.48 in 0–20 cm depth was higher in SRC than in cropland, but mostly not significant. Significant differences at pF 4.2 and others in the subsoil can be attributed to varying clay contents between the plots (*cf*. Tab. 1). The air capacity in the soil was significantly higher in 0–10 cm depth under SRC ('high') than in the cropland ('medium'), and also in 10–20 cm depth in SRC-W (Tab. 2). The proportion of plant-available water in the soil was significantly higher in 0–10 cm depth under SRC than in the cropland, while it was significantly lower in 20–30 cm depth.

The penetration resistance showed distinct changes with depth, with rather similar values in the topsoil of all plots and pronounced differences in deeper soil layers (Fig. 2). In the SRC plots, a gradual increase was observed down to 30 cm (SRC-P) or 40 cm (SRC-W) depth. The penetration resistance then remained constant for about 10 cm at around 2.2 MPa (SRC-P) or 2.9 MPa (SRC-W), and underneath decreased again at SRC-W, and continued to increase at SRC-P. In the cropland, the penetration resistance increased sharply at 22 cm (Cropland-W) or 26 cm (Cropland-P) depth and showed very distinct peaks around 40 cm depth, with values of 3.9 MPa (Cropland-W) or 4.3 MPa (Cropland-P), being significantly higher than the corresponding penetration resistance under SRC. Underneath, the penetration resistance in the cropland decreased again, remaining higher than that at the respective paired SRC plot.

The penetration resistance under cropland was distinctly higher than the assumed hydrostatic stress down to about 70 cm depth (Fig. 2), demonstrating that the soil was in a

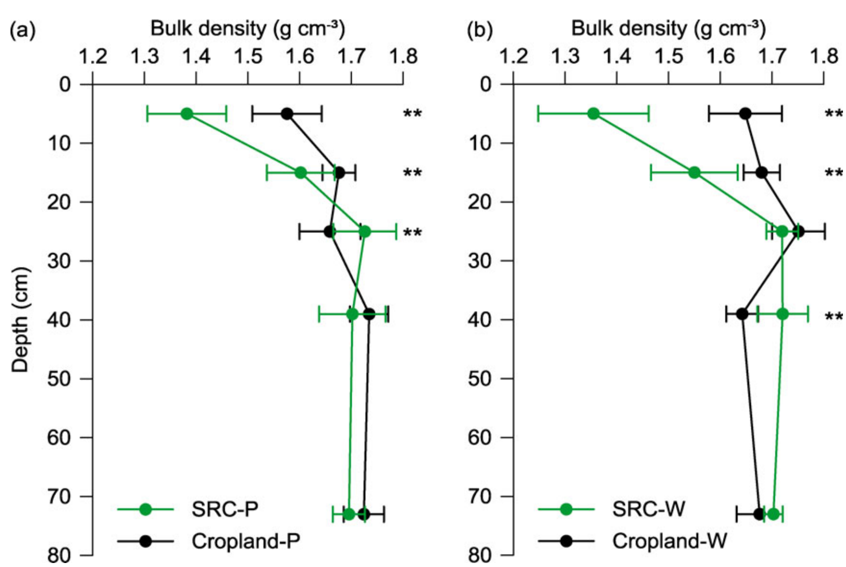

Figure 1: Bulk density after 18 years of short rotation coppice with (a) poplar (SRC-P) and (b) willow (SRC-W), and at paired references sites under cropland (Cropland-P, Cropland-W). Arithmetic mean ± one standard deviation, $n = 8$. Stars indicate significant difference between SRC and paired cropland at $p < 0.01$.

Table 2: Soil physical properties (arithmetic mean ± standard deviation, n = 8) after 18 years of short rotation coppice with poplar (SRC-P) and willow (SRC-W), and at paired reference sites under cropland (Cropland-P, Cropland-W). Θ: volumetric water content.[a]

Plot	Depth (cm)	$\Theta_{pF1.8}$ (%)	$\Theta_{pF2.48}$ (%)	$\Theta_{pF4.2}$ (%)	Total porosity (%)	Air capacity (%)	Plant- available water (%)
SRC-P	0–10	31.3 ± 0.8**	28.1 ± 0.9**	7.5 ± 0.3	47.8 ± 1.4**	16.6 ± 1.8*	23.8 ± 0.6**
	10–20	26.3 ± 1.2	24.2 ± 1.2	7.0 ± 0.7	39.5 ± 1.2*	13.2 ± 1.4	19.3 ± 1.0
	20–30	24.6 ± 0.3**	22.1 ± 0.3	8.4 ± 0.2**	34.9 ± 1.1	10.2 ± 1.0	16.3 ± 0.3**
	35–45	25.3 ± 1.0**	22.5 ± 1.0**	9.1 ± 0.5**	35.8 ± 1.1	10.5 ± 1.3**	16.2 ± 0.4**
	70–80	23.3 ± 0.4**	18.3 ± 0.3**	6.4 ± 0.2**	36.3 ± 0.7	13.0 ± 0.9**	16.9 ± 0.4**
Cropland-P	0–10	27.8 ± 0.8	24.4 ± 0.6	7.2 ± 0.4	40.5 ± 1.2	12.8 ± 1.6	20.6 ± 0.7
	10–20	25.1 ± 1.1	22.3 ± 0.9	7.0 ± 0.0	36.8 ± 0.6	11.6 ± 1.4	18.1 ± 0.4
	20–30	28.3 ± 0.5	23.3 ± 0.7	6.9 ± 0.0	37.4 ± 1.1	9.2 ± 1.3	21.3 ± 0.2
	35–45	29.4 ± 0.6	27.4 ± 0.7	16.5 ± 0.6	34.6 ± 0.7	5.2 ± 1.3	12.9 ± 0.7
	70–80	28.5 ± 0.3	26.4 ± 0.3	17.1 ± 0.0	34.9 ± 0.7	6.4 ± 0.9	11.5 ± 0.3
SRC-W	0–10	28.9 ± 0.7	26.7 ± 0.5**	5.5 ± 0.1	48.9 ± 1.1**	20.0 ± 1.4**	23.4 ± 0.6*
	10–20	26.2 ± 0.6	23.9 ± 0.7	5.4 ± 0.0**	41.5 ± 1.6**	17.0 ± 1.4**	20.7 ± 0.6
	20–30	24.0 ± 0.7	21.2 ± 0.6	6.4 ± 0.1	35.1 ± 0.6	11.1 ± 0.9	17.6 ± 0.3*
	35–45	23.4 ± 0.5	21.1 ± 0.6	6.1 ± 0.1**	35.1 ± 0.9**	11.7 ± 0.9*	17.3 ± 0.7
	70–80	25.6 ± 0.8**	23.4 ± 1.1**	9.4 ± 0.2**	35.8 ± 0.4	10.1 ± 0.7**	16.2 ± 0.5
Cropland-W	0–10	28.3 ± 0.6	23.0 ± 0.8	6.5 ± 0.6	37.8 ± 1.3	9.5 ± 1.6	21.8 ± 0.5
	10–20	26.5 ± 0.8	21.8 ± 0.7	6.6 ± 0.4	36.6 ± 0.7	10.1 ± 0.9	20.0 ± 0.6
	20–30	25.8 ± 1.2	21.9 ± 1.5	6.2 ± 0.0	33.9 ± 0.9	8.1 ± 1.0	19.6 ± 0.9
	35–45	23.3 ± 0.8	19.4 ± 1.0	5.2 ± 0.1	38.0 ± 0.6	14.8 ± 1.1	18.1 ± 0.5
	70–80	22.3 ± 1.2	16.7 ± 2.4	4.4 ± 0.1	36.8 ± 0.8	14.5 ± 1.1	17.9 ± 1.1

[a]*Significant difference between SRC and paired reference site under cropland at p < 0.05; **significant difference between SRC and paired reference site under cropland at p < 0.01.

precompacted state. The stresses at rest coefficient K_0 peaked in about 30–40 cm depth with values of 2.6–2.8 MPa and 2.2–2.5 MPa in Cropland-W and Cropland-P, respectively. Below 70 cm, K_0 was ≤ 1, indicating normal compaction. In the SRC, in contrast, K_0 decreased gradually with depth, and values in 30–40 cm depth amounted to only 1.8–2.1 MPa (SRC-W) and 1.1–1.5 MPa (SRC-P). Normal compaction was reached already at 50–60 cm depth.

The WDPT measured on soil samples both from SRC and cropland never exceeded 5 s. All samples thus fell within the class 'wettable' according to *Dekker* and *Ritsema* (2000), and no water repellency was detected.

3.2 Flow patterns

The dye tracer images from both SRC plots could be divided into three parts (Fig. 3a). In the upper 10–15 cm, the staining was nearly continuous with a dye coverage of > 95%; the UID was 17 and 21 cm in SRC-P and SRC-W, respectively. Underneath, a high and uniform dye coverage was observed down to 40 cm depth. Below 40 cm depth, the dye coverage decreased sharply in SRC-P (Fig. 3d), while the transition was more gradual in SRC-W. Dye was transported down to the bottom of both profiles in many pathways. These were constituted by roots channels and earthworm burrows, with root channels being most abundant.

Under cropland, the staining was rather homogeneous and uniform down to 40–45 cm depth, corresponding to the ploughing horizon. In the experiments from 2011, dye infiltrated only down to a depth of a few centimetres in one half of the profiles; the reason was a former machinery track. Dye coverage was higher than under SRC in the middle of the profile, but lower in the bottom of the profile. Also under arable use, flow pathways reached down to the bottom of the profile. Flow pathways were less and made up of earthworm burrows. The PFF was significantly smaller under SRC, while LI was significantly higher.

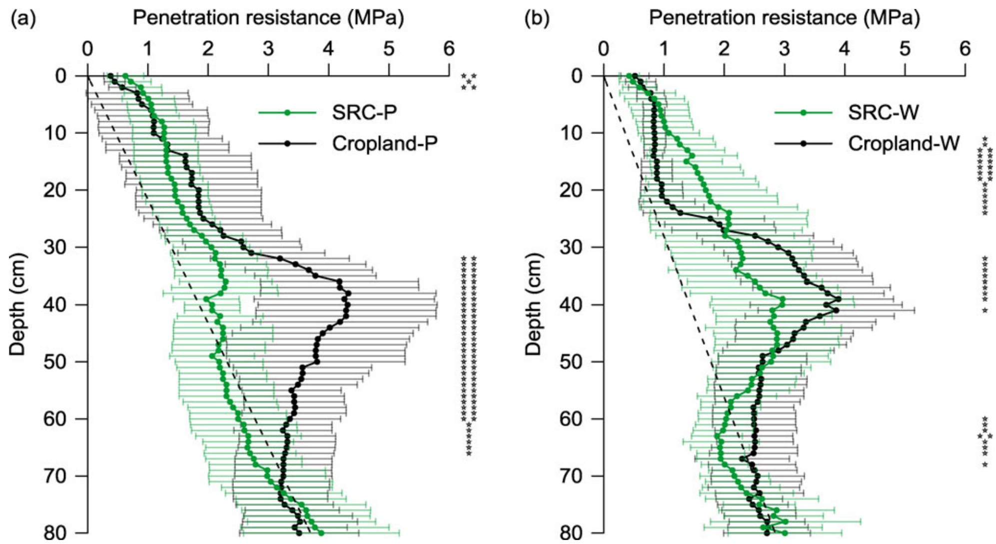

Figure 2: Penetration resistance measured with a penetrologger under short rotation coppice with (a) poplar (SRC-P) and (b) willow (SRC-W), and at paired reference sites under cropland (Cropland-P, Cropland-W). Arithmetic mean ± one standard deviation, n = 15. Stars mark significant differences between treatments at p < 0.05 (*) or p < 0.01 (**). The dashed line shows the assumed hydrostatic stress distribution. Soil depths with penetration resistance values greater than this line are in a precompacted state.

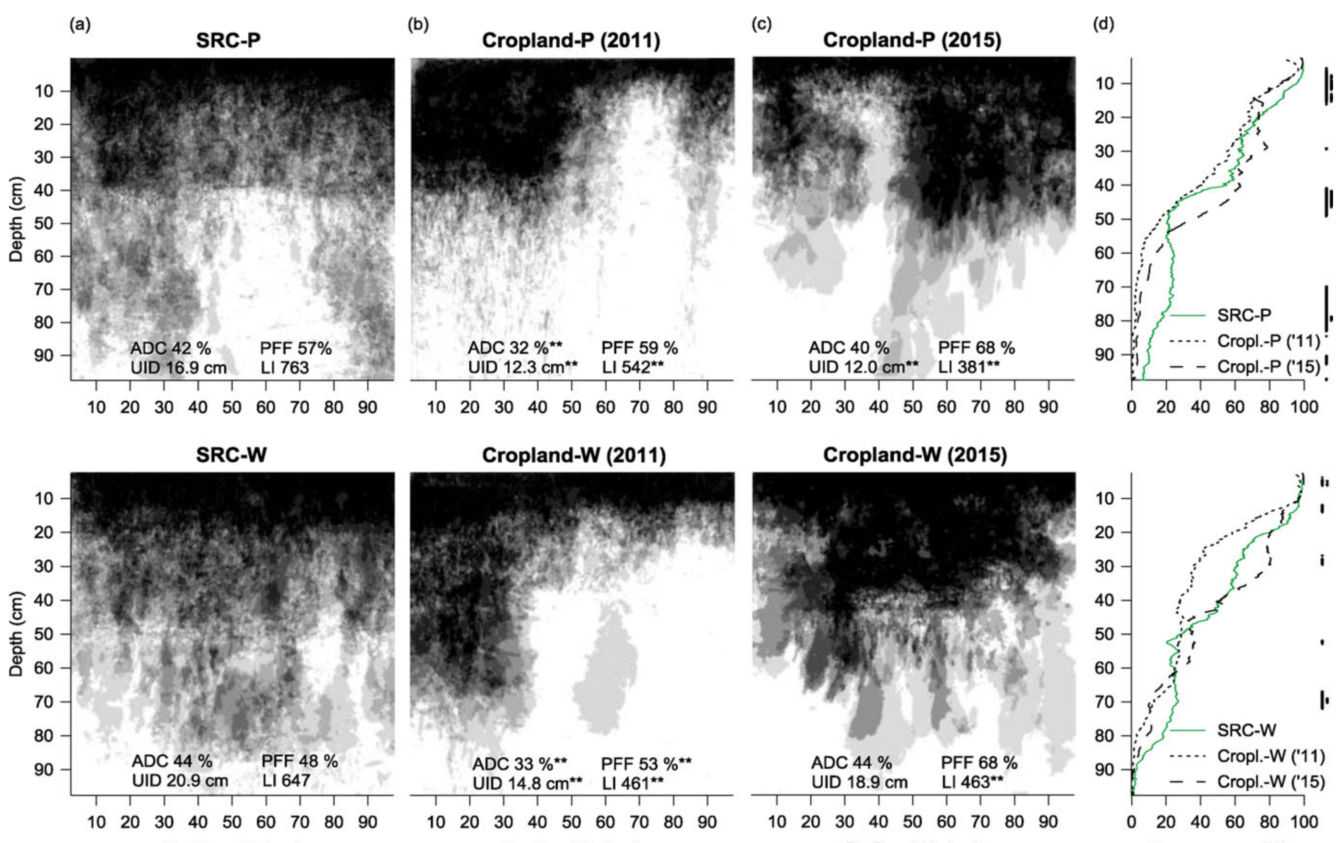

Figure 3: Spatial distribution of dye in super-imposed images (n = 7) of soil profiles (a) after 18 years of short rotation coppice with poplar (SRC-P) and willow (SRC-W) and (b), (c) at paired sites under cropland (Cropland-P, Cropland-W). Black colour indicates that each single image was stained, white colour indicates that no image was stained. ACD: average dye coverage, PFF: preferential flow fraction, UID: uniform infiltration depth, LI: length index. (d) Average dye coverage with depth for the same images.

4 Discussion

4.1 Impact of short rotation coppice on bulk density

The bulk density was significantly lower in the SRC's topsoil than in that of the cropland reference. Upon establishment of the SRC, the bulk density of the topsoil was 1.65 g cm^{-3}, which is similar to the values 1.64/1.69 g cm^{-3} (average over 0–30 cm under SRC-P/SRC-W) measured at the cropland reference sites in 2011. These values were measured 10 months after the last tillage operation, and it can be assumed that the soil particles had settled again and the bulk density stabilized (as described in *Kool* et al., 2019), making the value suitable for comparison. Bulk densities measured in the topsoil of another, 24-year-old SRC near Rostock showed even lower values (and larger differences to the cropland reference site), which we attribute to a longer coppice age and a higher planting density compared to Gülzow (*Giggel*, 2010). These results show that long-term SRC with poplar and willow causes a loosening of the topsoil in substrates typical for northern Germany, which is a positive effect for biomass production, since enhanced bulk densities and penetration resistance have been shown to result in a significant reduction of biomass production in SRC (*Souch* et al., 2004). Decreasing bulk densities have also been reported upon afforestation with hybrid aspen, birch, pine and spruce on former agricultural land (*Messing* et al., 1997; *Wall* and *Hytönen*, 2005; *Olszewska* and *Smal*, 2008).

Upon conversion of an arable field to SRC, two opposing trends are activated: on the one hand, abandoning tillage is often associated with soil compaction and decreasing porosity in the topsoil (*Tebrügge* and *Düring*, 1999; *Moret* and *Arrúe*, 2007). The reason is the natural settlement of the soil with a progressive alignment of the soil particles. On the other hand, root physiology, root distributions, and root biomass quality are different from agricultural crops and tend to increase porosity under trees (*Messing* et al., 1997; *Gill* and *Jackson*, 2000). Trees can have up to five times more biomass than cereals in the uppermost 40 cm of the soil (*Messing* et al., 1997), and typically 75–95% of willow and poplar fine roots occur within the topsoil (*Crow* and *Houston*, 2004) with a lifespan of days to months (*Block* et al., 2006).

In addition to the loosening effect of the tree roots, bulk densities in the SRC topsoils were reduced due to the accumulation of organic carbon originating from leaf litter inputs and tree rooting (Tab. 1; see also *Kahle* et al., 2010). The SOC content was distinctly higher under SRC than under cropland in the upper soil layer (0–10 cm).

The bulk density and soil structure are also influenced by the colonisation with invertebrates, for which reduced soil tillage, absence of pesticides, and accumulation of litter and soil organic matter are beneficial effects of SRC. An increase in abundance and mass of earthworms, woodlice, and harvestmen under SRC compared to adjacent arable fields has been observed (*Makeschin*, 1994), and population density in general rises quickly after reducing tillage (*Ojha* and *Devkota*, 2014). At the study site, the abundance, biomass and diversity of earthworms were significantly higher under SRC than under cropland (*Saggau*, 2011). Furthermore, the proportion of adult (80 individues m^{-2}) to juvenile individuals (283 individuals m^{-2}) was shifted towards juveniles, proving a high reproduction rate of the earthworms. These changes under SRC illustrate a gradual biogenic regeneration of the soil. Earthworms do not only increase the porosity of the soil, but also create biogenic channels, enhancing water flow, aeration, and root development (*Emmerling* et al., 2002). They entail an increasing heterogeneity of soil physical properties, which is reflected by the larger standard deviation of the bulk density at the SRC compared to the cropland reference sites. It has been shown that SRC provides a more attractive habitat than cropped land also for small mammals, as evidenced by overall density, species richness and composition, with older coppice being most attractive (*Christian* et al., 1998; *Rowe* et al., 2009).

In the subsoil, bulk densities were similar but high under both types of land use due to the glacial till parent material. Roots of poplar and willow have been observed to reach down to > 110 cm depth, but the number of roots was obviously too low to have an influence on bulk density. It must be noted that undisturbed soil samples could not be taken in areas of active, large roots.

4.2 Impact of short rotation coppice on water retention

Distinct effects of the SRC were also observed on the water retention, with an increase in plant-available water in the upper soil layer. We assume that this increase is caused by the accumulation of soil organic matter (+ 0.6% SOC in 0–10 cm depth; Tab. 1), empty fine root channels and soil loosening as well as a higher proportion of water-stable soil aggregates, which were found in the upper soil of the SRC and have the capacity to store water (*Kahle* et al., 2013). These factors also reduce the bulk density. The linkage between bulk density and hydraulic soil properties is not straightforward, though, since bulk density only gives an average value of mass per volume, but no information on where in the volume the mass is located. The availability of soil water for plant roots might be restricted by the tortuosity and connectivity of the pore network. However, the dye tracer images show rather patches of dye than narrow pathways, indicating that the dye spread quite easily form the macropores (as main flow paths) into the soil matrix. We deduce that the movement of water towards plant roots is not restricted, and existing soil water is actually available for plants in the upper 70–80 cm of the soil.

The increase of the plant-available water content is a positive effect of SRC, since the yield of willow, poplar and hybrid aspen has been shown to be positively correlated with the available water content (*Tullus* et al., 2010; *Petzold* et al., 2011; *Lutter* et al., 2017). In contrast to our results, *Olszewska* and *Smal* (2008) found no significant difference between the water retention capacity of long-term afforested and agricultural soils, possibly due to a high sand content (> 90%).

4.3 Evidence for loosening of plough pan

The highest penetration resistance was observed at 30–45 cm depth under cropland and was located at the basis of the ploughing horizon. It reflects a compacted plough pan. In agreement to this, the calculated stresses at rest show distinct peaks in the same depths, indicating a strong precompaction of the soil. Overall, the cropland soil was in a precompacted state ($K_0 > 1$) down to 70 cm depth as a result of the mechanical soil loading. Under SRC, the penetration resistance in 30–45 cm depth also showed an increase compared to the topsoil, but exhibited much lower values. The hydrostatic stresses decrease gradually with depth under SRC and show a strongly reduced precompaction compared to the cropland. These findings indicate that long-term SRC is able to break a plough pan compaction as a result of the long absence of ploughing, numerous tree root channels in the soil, soil organic matter accumulation, and the increased earthworm activity. This is a positive evolution, since a penetration resistance beyond 2 MPa has been shown to slow down root elongation considerably (*Bengough* et al., 2011).

A previous study on Cambisol (loamy texture) and Gleysol (sandy-clayey texture) soils had not found any evidence for a perforation of the plough pan four years after SRC establishment on a formerly cropped field (*Makeschin*, 1994). These diverging results show that the effect of roots on soil structure depends on the stand age of the SRC and other site-specific characteristics. The perforation of compacted soil layers is determined by soil characteristics, such as degree, thickness and depth of compacted soil layers, mechanical properties, and water and nutrient distribution, by plant characteristics like diameter and tortuosity of the roots, and by earthworm population (*Dexter*, 1978; *Marinissen*, 1992; *Bengough* et al., 2006).

4.4 Differences in flow patterns between short rotation coppice and conventional cropping

Dye tracer experiments revealed different stain patterns under SRC and conventional cropping. At the SRC, the nearly complete dye coverage (significantly higher than at the cropland) in the upper 10–15 cm reflects a uniform infiltration and corresponds well to the low bulk density and large proportion of coarse pores detected in these depths, resulting in a deeper UID than in the cropland. Underneath, the high dye coverage down to 40 cm depth corresponds to the former ploughing layer with a rather low bulk density and penetration resistance. It was surprising how sharp the plough pan, originating from the former agricultural use, was visible especially under poplar. Figure 3a demonstrates that the plough pan still influences water and solute fluxes in the soil profile even after 18 years of SRC without any tillage operations, and even though the penetration resistance indicated a loosening of the plough pan. Through and underneath the plough pan, the dye was transported mainly in macropores, partly down to the profile basis at 1 m depth. Both root channels of coarse roots and earthworm burrows were observed. Even though the deep roots observed under SRC did not significantly influence bulk density, we assume that they contributed to solute transport und explain—together with the higher abundance of earthworms (*Saggau*, 2011)—the higher dye coverage in the SRC below 50–60 cm depth. Trees have a larger proportion of coarse roots than conventional crops. Coarse roots are lignified and have a decelerated root turnover (*Gill* and *Jackson*, 2000) and leave behind stable voids, which then can act as pathways for solute transport.

The PFF was distinctly smaller under SRC than under cropland, which can be attributed to the fact that the UID was deeper there and, thus, a shorter part of the profile dominated by preferential flow. The LI, in contrast, indicates that preferential flow was significantly higher under SRC, reflecting that flow paths were more heterogeneous.

The experiments were conducted under ponding conditions. For the initiation of macropore flow, a water content near saturation is required, either at the soil surface or in a deeper soil horizon (*Weiler* and *Naef*, 2003). Macropore flow was initiated near the soil surface in the SRC, and macropores were continuous from the soil surface to the subsoil. Under cropland, macropore flow was initiated on top of the plough pan, with presumably reduced hydraulic conductivity, where stained macropores started and continued down to the subsoil. Macropores were disrupted in the topsoil as a result of tillage operations. Upon high-intensity rainfall events, or when rainfall occurs at high soil water contents, the topsoil may become water-saturated and a preferential transport of dissolved chemicals to deeper soil layers can thus be expected in all profiles. It can be anticipated that—upon the same input—the solute concentration in the percolation water will be higher under SRC than under cropland because of preferential flow.

The generation of preferential flow might be enhanced by water repellency. The occurrence of water repellancy was hypothesized because *Jandl* et al. (2012) had found higher concentrations of long C-chain aliphatic lipids, sterols and suberins under SRC than under annual arable crops at the study site in 0–10 cm depth. This change in the molecular composition and stability of the soil organic matter, however, has not entailed hydrophobicity according to our results.

5 Conclusions

Combined investigations of undisturbed soil samples and solute transport patterns indicate a distinct impact of long-term SRC on soil physical properties. The decreased bulk density found in the topsoil of SRC compared to conventional arable use promotes the plant-available water supply, which possibly enhances the biomass production of the SRC. The increased porosity furthermore improves the living conditions for soil biota.

The improved water retention capacity is expected to be of advantage also after reconversion of the SRC to annual crops. The same applies to the observed loosening of the plough pan, which reduces the mechanical resistance to root penetration also for future crops. Dye tracer patterns revealed that the continuity of macropores in the soil profile and the potential for a deep transport of solutes is higher under SRC. Special caution should therefore be given when agrochemicals are applied both upon reconversion and on marginal sites. A lasting challenge is the sustainability of the positive evolution after reconversion of SRC into arable land and therefore the selection of suitable tillage strategies for the reconversion.

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